

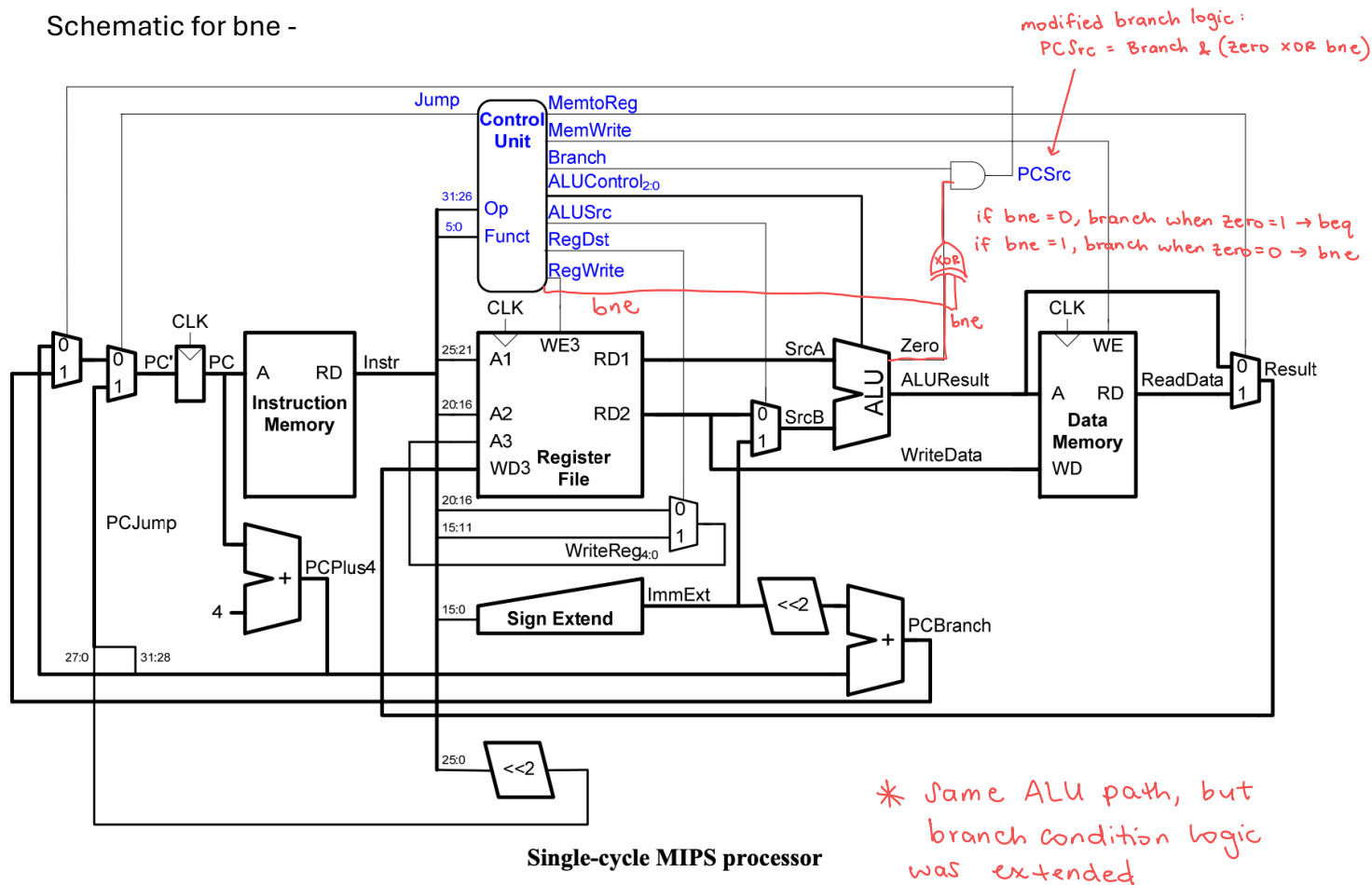
Lab 6: MIPS single-cycle CPU

Part B:

New Instructions Implemented

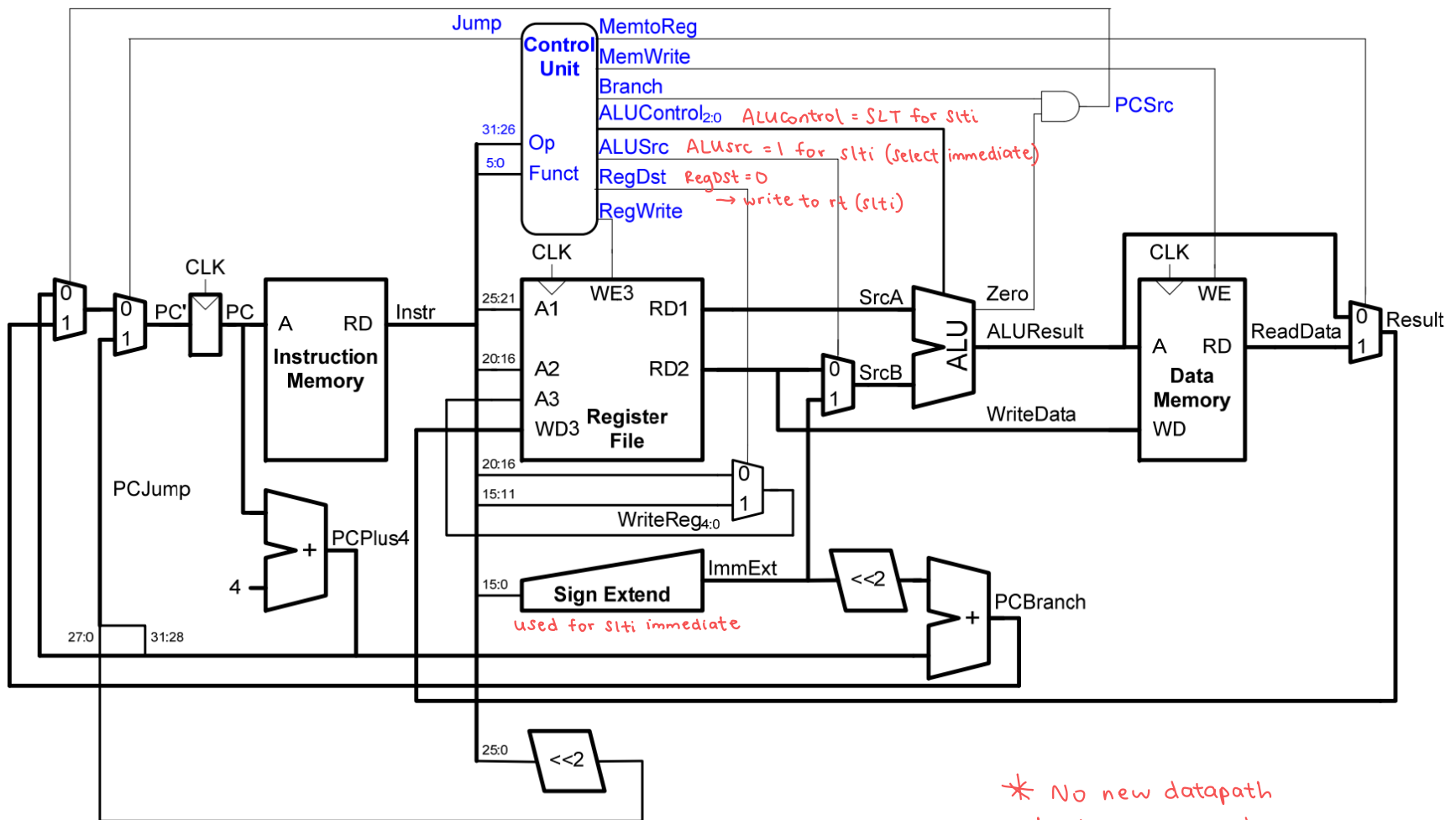
- bne with opcode 000101
 - bne rs, rt, imm: branch when rs != rt

Schematic for bne -



- slti with opcode 001010
 - slti rt, rs, imm: set rt = 1 if rs < imm using signed compare, else 0

Schematic for slti -



Single-cycle MIPS processor

Filled in for bne and slti:

Extended functionality. Main Decoder:

Instruction	Op _{5:0}	RegWrite	RegDst	AluSrc	Branch	MemWrite	MemtoReg	ALUOp _{1:0}	Jump			
R-type	000000	1	1	0	0	0	0	10	0			
lw	100011	1	0	1	0	0	1	00	0			
sw	101011	0	X	1	0	1	X	00	0			
beq	000100	0	X	0	1	0	X	01	0			
addi	001000	1	0	1	0	0	0	00	0			
j	000010	0	X	X	X	0	X	XX	1			
bne	000101	0	x	0	1	0	x	01	0			
slti	001010	1	0	1	0	0	0	11	0			

Extended functionality. ALU Decoder:

ALUOp _{1:0}	Meaning
00	Add
01	Subtract
10	Look at funct field
11	slt